

CS 205 – Professional and Societal Issues in Computing

Peer Review 1 -> Monday, April 6th

Peer Review 2 -> Monday, April 13th

▶ Due Thursday, April 23rd, 23:59 PST ▶

1 Introduction

The ethical question at the heart of this exercise is the following:

“Is it ethical for software engineers to use modern generative AI applications to generate code that is used, in whole or in part, for commercial applications?”

This is a very broad question! To give more shape to this question, let's boil down what I mean:

- Software Engineers

This could be a whole team of 100+ engineers working on enterprise-level software suites, a mid-size team working on an indie game, or a single engineer working on freelance phone apps or web-dev.

- Modern Generative AI Applications

About what it sounds like. Let's mostly consider current/recent versions of GenAI models, made within the last year or two. You can consider older models as part of your analysis/research, but the code we're debating for the question above would be from relatively recent GenAI.

- In whole or in part

Consider that code can be generated from an LLM, and then edited so that it no longer resembles its initial form (like the Ship of Theseus)[https://en.wikipedia.org/wiki/Ship_of_Theseus]. You should take into account that a software engineer could use the generated code without changing it at all; or it could be changed somewhat; or it could be changed almost completely! Maybe only variable names are changed and compile errors removed... but any of the above are possible.

- Commercial Applications

Meaning that the software in question will be used for some kind of commercial application. It doesn't necessarily need to be sold; it could be in-house software used for other purposes (for example, it could be used to generate a nice web UI/UX portal for using an in-house database), or licensed to a third party for some other use.

I also didn't specify whether the code is proprietary or open source. That will be part of your consideration and research!

2 Report Structure

- **Background Information:** Set up overall problem/issue that this paper is addressing. Who are the stakeholders, or concerned parties, in the current scenario? Identify them and present their roles in this scenario.
- **Current event analysis:** What is the current state of generative AI in regards to code generation? Describe current examples of benefits, limitations, and/or misuse, related to Generative AI tools.
- **Technical Background:** Provide insight into the technical process behind how AI models generate code. This breakdown doesn't need to be as thorough as the tech section from the Pegasus Paper, but should give an adequate description of how AI models generate code output, given a text input prompt. This section (not counting diagrams, if you elect to use any) should be roughly 1.5 to 2 pages.
- **Ethical Analysis:** Present your analysis of the central premise, and a formal argument to support or refute the ethical question at the heart of the issue. Remember to use one or more of the frameworks discussed in class. You may combine concepts / theorems / frameworks from multiple theories to support your argument (you do not need to use EVERY framework):
 - Kantianism
 - Rule Utilitarianism
 - Virtue Ethics
 - Social Contract Theory

- **Relevance and conclusion:** Identify why the central question concerning these events is relevant (in particular) to the discussion in this course, and (in general) to the larger society. Is there more information that you can provide, as a practitioner in computer science, towards the ethical responsibilities of AI-based tools used in “generating” code? What are some technical safeguards that you can add to the AI framework to make it robust against such misuse? *How has your research shaped your understanding of issues underlying generative AI tools? Are there any significant takeaways that you can apply to your projects/experience in DigiPen?*

Your report should be submitted using the ACM template, and be 5-7 pages in length (5 pages minimum), excluding the list of references. **You are required to search for/provide at least 10 distinct, reliable sources and cite them with appropriate reference format.**

Reminder: WIKIPEDIA IS NOT A SOURCE! It is a collection of references that points to appropriate sources. If you are going to use Wikipedia, you need to visit the sources cited in the Wiki article, and reference those instead.

Rough recommended page length per section:

Intro + Before/Current/After events	->	1.5 - 2 pages
Technical Details	->	1.5 – 2 pages
Ethical Analysis + Conclusion	->	1.5 – 2.5 pages
Beyond the Superficial	->	0.5 pages
References	->	Does NOT COUNT toward page count
Total Pages	->	5 - 7 pages

Can I use Generative AI for this paper?

Since the topic of this paper is code generation, you are allowed to use generative AI to generate code. The purpose of doing so should be to use any output as specific examples of how AI models can generate code, as a means for strengthening the arguments in your paper. **This is the ONLY valid use of generative AI for this paper!**

It should go without saying that you should not use generative AI as a paper-writing aid, in any way.

3 Grading Criteria and Rubric

Criteria

Item	Weight
Understanding the current state of Generative AI & providing background information	25%
Analysis of the architecture of modern AI models, with respect to code generation	15%
Ethical Analysis (answer to the ethical question at the heart of the paper)	20%
Conclusion: Relevance to individuals in the field of computing (including self)	10%
Writing style and grammatical errors	10%
Presentation style / tools (diagrams, layout/structure of paper)	10%
Using correct attribution to references/sources	10%

Rubric

Item	Exemplary	Satisfactory	Developing	Unsatisfactory
Understanding the topic & providing background information	Clear, concise introduction to the topic, that sets up the analysis in a logical fashion	Introduction may be slightly unclear in setting up the goals of the paper.	Introduction does not clarify the goals of the paper.	Introduction too long/short, or not written at all.
Using correct attribution to references/sources	References are cited in the text and provided at the end of the paper in appropriate order.	N/A	N/A	References are not cited, or not provided in appropriate order.
Analysis using Workable Ethical Theories	The parameter / hypothesis is correctly identified and listed.	Parameters are listed, but the formulation of the problem statement is unclear.	Partial representation of the parameters in the paper.	Neither the parameters nor the hypothesis is listed in the paper.
Providing an assessment or conclusion for the topic	Clear, concise conclusion to the topic, that follows from the discussion in a logical fashion	Conclusion is unclear in setting up any tangible action to implement.	Conclusion is partially provided, or is unclear in the goal.	Conclusion is too short or not provided at all
Writing style/format, grammatical errors, and presentation style/tools	No grammatical errors, and the text is presented in a clear format, sufficient supporting diagrams/material	Some grammatical errors, and the text is unclear, and/or few supporting diagrams/material	N/A	Grammatical errors and no text provided
Tech Analysis of AI Models and any example output	Generative AI frameworks are described in sufficient detail for a computer science audience. Any generative AI output (example code, etc) is analyzed in some detail.	AI models are described in some detail, but either missing key points or too broad to appeal to a computer science audience.		Inner workings of AI models are not discussed.